Specifying and Estimating Confirmatory Factor Models

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Table of contents

Chapter box

Five themes.
A restriction is a claim. Confirmatory analysis differs from exploratory analysis in what it fixes, not in the equation it uses. Every zero is a statement about the moments that could turn out to be false.
Fixed, free and constrained are three different statuses. A diagram shows which relations exist; it does not show which are estimated, which are held at a value, and which are tied to another.
A latent variable needs a metric before it has coefficients. Marker scaling and fixed-factor scaling give the same implied covariance and different printed numbers.
Counting is necessary and not sufficient. Moments minus free parameters gives degrees of freedom; identification asks the separate question of whether the moments determine the parameters.
Estimation minimizes a stated discrepancy. Convergence, admissibility and a well-conditioned solution are read before any coefficient is interpreted.
Prerequisites. The covariance decomposition, factor scale and orientation, and pattern coefficients from the preceding chapters, together with the candidate model fixed at the end of the exploratory analysis.
Records. A neutral six-indicator model makes the algebra visible; a four-factor topology tests diagram-to-syntax transfer; a separate synthetic example supports compact comparisons. The twelve-indicator candidate is estimated on the confirmation sample, untouched by the development analysis.
Scope. We specify, identify, estimate and check. Convergence and admissibility are settled here; overall model adequacy is not assessed, and no fit index is interpreted.

1 From exploratory structure to restricted CFA

1.1 The change is in the restrictions

The common-factor equation has not changed:

$$\mathbf{y}_i = \nu + \Lambda\eta_i + \epsilon_i, \qquad \Sigma(\vartheta) = \Lambda\Phi\Lambda^{\mathsf{T}} + \Theta.$$

What changes in confirmatory factor analysis is how the parameters are restricted before estimation. A CFA might assign each indicator to one factor, allow a few specified secondary loadings, constrain selected loadings to equality, or specify a local residual covariance. The model then has testable implications beyond the number of factors alone.

Here ϑ denotes the vector of free parameters; Θ denotes the residual covariance matrix. They are not two notations for the same object. For p indicators and m factors, Λ is $p \times m$, Φ is $m \times m$, and Θ is $p \times p$. As throughout Part Two, $\Phi = \mathrm{Var}(\eta)$.

Question	EFA in Chapter 2	CFA in this chapter
Which loadings are estimated?	All, before choosing an orientation	Only the declared free relations
How is orientation handled?	Rotation and subsequent labels	Loading restrictions and factor metrics
What determines the candidate?	Development evidence plus content	A fixed specification, with named rivals
What data are being used?	Sample A for development	Sample B for estimation of that candidate
What has not been established?	Unique substantive interpretation	Adequacy, validity, or causality

"Confirmatory" describes restrictions and analytic history, not a guaranteed favorable conclusion. A CFA estimated after repeated same-data changes is still mathematically a CFA, but the changed specification is data-developed. Conversely, a prespecified cross-loading does not make a model nonconfirmatory merely because it departs from independent-clusters simple structure.

1.2 Missing arrows have statistical consequences

An absent loading arrow generally means a coefficient fixed to zero. It does not mean that the indicator and factor are uncorrelated. Under correlated factors, an indicator can correlate with a factor on which it has no direct loading. Chapter 2's pattern–structure distinction therefore remains essential.

An absent residual covariance means that two residuals are uncorrelated under the model. That is a moment restriction. To assert conditional independence of indicators given the factors, we additionally need an appropriate conditional distribution or factorization; in a jointly normal linear model, diagonal residual covariance supplies that stronger property. Zero covariance alone does not generally imply independence.

A factor-to-indicator arrow specifies a loading relation. Interpreting the arrow causally requires design and substantive assumptions beyond the covariance equation. Good fit cannot supply temporal precedence or exclude every alternative explanation.

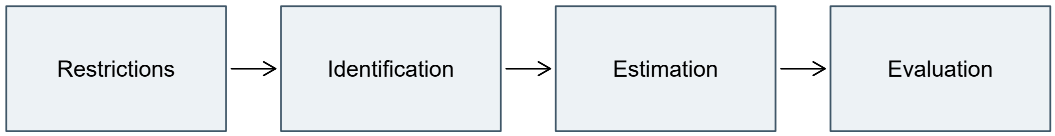


Figure 1: Specification, recoverability, and numerical estimation are distinct tasks; evaluating the fitted record is a fourth.

2 Write one model four ways

2.1 A verbal specification

Begin with six continuous indicators, $Y_1, \dots, Y_6$, and two correlated factors. The first three indicators load only on η_1; the last three load only on η_2. All six residual variances are free and all residual covariances are zero. Factors and residuals are uncorrelated. Use Y_1 and Y_4 as markers, fixing their loadings to one. The factor variances and their covariance remain free.

The labels are intentionally neutral. This compact model is adapted from a two-factor, six-indicator teaching structure, not extracted from the SLE items. It lets us follow each restriction without introducing a second substantive measurement story.

2.2 Six scalar equations

With zero latent intercepts and zero-mean residuals,

$$\begin{aligned}
Y_{i1} &= \nu_1 + \eta_{i1} + \epsilon_{i1}, & Y_{i4} &= \nu_4 + \eta_{i2} + \epsilon_{i4}, \\
Y_{i2} &= \nu_2 + \lambda_{21}\eta_{i1} + \epsilon_{i2}, & Y_{i5} &= \nu_5 + \lambda_{52}\eta_{i2} + \epsilon_{i5}, \\
Y_{i3} &= \nu_3 + \lambda_{31}\eta_{i1} + \epsilon_{i3}, & Y_{i6} &= \nu_6 + \lambda_{62}\eta_{i2} + \epsilon_{i6}.
\end{aligned}$$

The subscript i denotes a person; the first subscript of λ_{ja} identifies the indicator, and the second identifies the factor. Writing all six equations is useful once: it prevents a compact diagram from hiding a cross-loading or a mistaken marker. Later, matrix notation avoids repeating the same structure.

2.3 Three parameter matrices

The same model is

$$\Lambda = \begin{bmatrix} 1 & 0 \\ \lambda_{21} & 0 \\ \lambda_{31} & 0 \\ 0 & 1 \\ 0 & \lambda_{52} \\ 0 & \lambda_{62} \end{bmatrix}, \qquad \Phi = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{12} & \phi_{22} \end{bmatrix}, \qquad \Theta = \mathrm{diag}(\theta_{11}, \dots, \theta_{66}).$$

There are twelve loading slots, not twelve freely estimated loadings: four are free, two are fixed at one for scaling, and six are fixed at zero. Symmetry means ϕ_{12} and ϕ_{21} are the same covariance, not two independent parameters. Likewise, a freely estimated residual covariance would occupy two symmetric matrix entries but add only one free parameter.

The implied mean vector is $\mu = \nu + \Lambda\alpha$. Here $\alpha = \mathbf{0}$, so $\mu = \nu$. The latent intercept equals the latent mean because this CFA has no structural paths or latent predictors. This convention will need an explicit distinction between intercepts and marginal means once structural regressions enter.

2.4 A diagram and executable syntax

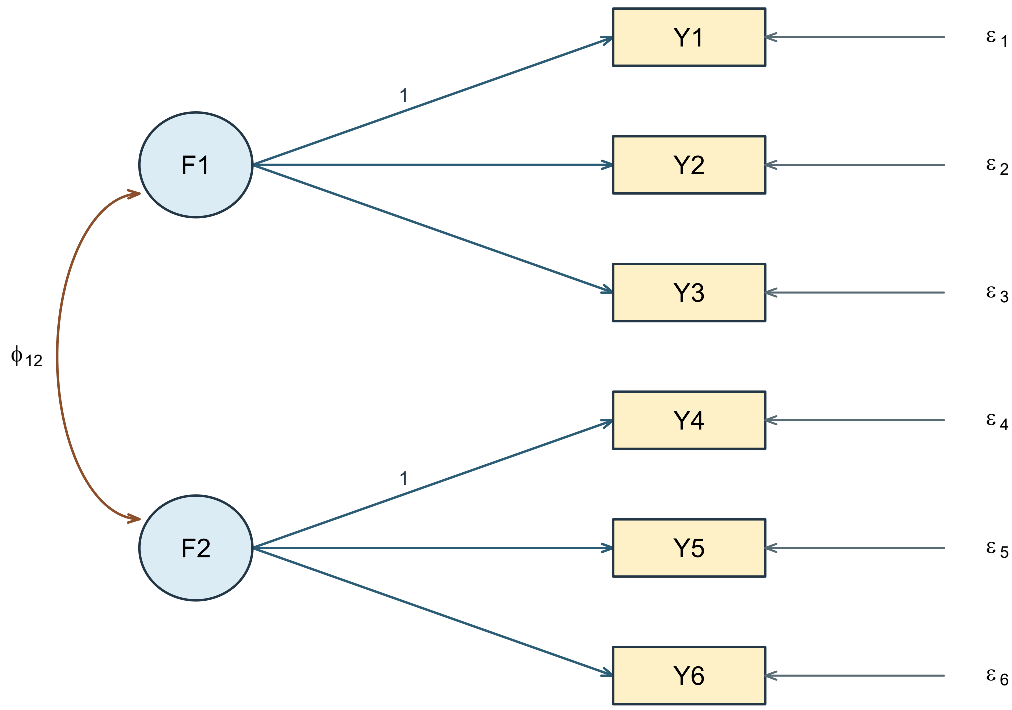


Figure 2: Marker-scaled six-indicator CFA. Residual variances are free; residual covariances and unshown loadings are zero. Intercepts are omitted from this covariance diagram.

In R, the `lavaan` package represents factor loadings with `=~` and variances or covariances with `~~`. The following is the marker-scaled specification:

```
model6 <- '
  F1 =~ 1*Y1 + Y2 + Y3
  F2 =~ 1*Y4 + Y5 + Y6
  F1 ~~ F2
'
```

The `cfa()` convenience function supplies free indicator residual variances and fixes unmentioned residual covariances to zero in this model. Defaults must still be checked; a short syntax string is not a complete parameter ledger. The official CFA tutorial is a useful orientation to the specify–fit–extract workflow.

Representation	What to check
Words	Indicator assignments, residual assumptions, metric, and means
Scalar equations	Every indicator has exactly the stated loading terms
Matrices	All zero, fixed, and shared entries are explicit
Diagram and syntax	Every drawn or coded relation matches the same ledger

If two representations disagree, no amount of numerical convergence can resolve which model the researcher intended. Fix the specification before interpreting output.

2.5 Transfer the grammar to four factors

A larger diagram should be read with the same ledger rather than as a new kind of model. As a published-reading exercise, reconstruct the topology of Figure 3 in Chou & Chou (2021): four correlated factors—self-efficacy, school support, privacy concerns, and technostress—each measured by four indicators. The reconstruction below records the displayed assignments and factor relations; it does not reproduce the paper's estimated coefficients or imply that every software default used in the original analysis is known.

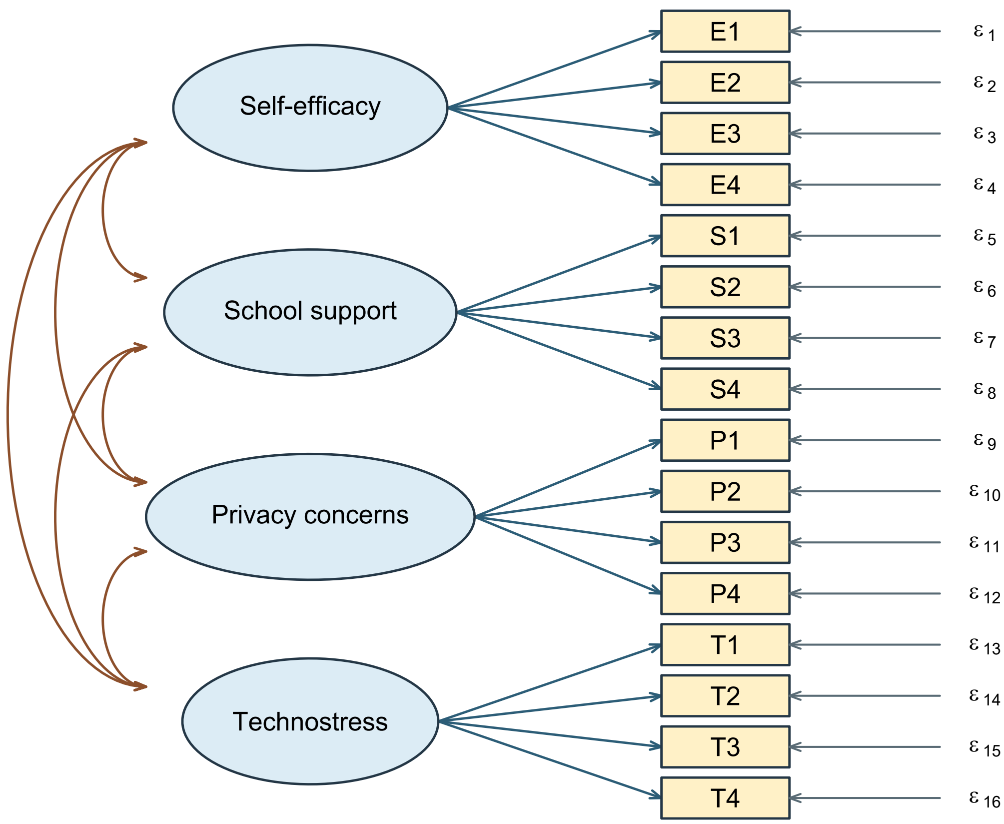


Figure 3: Four-factor, sixteen-indicator CFA reconstructed as a diagram-reading transfer. All six factor correlations are free; fixed-factor scaling is used for the count.

Under fixed-factor scaling, a direct `lavaan` translation is

```
model4 <- '
  SelfEfficacy =~ E1 + E2 + E3 + E4
  SchoolSupport =~ S1 + S2 + S3 + S4
  Privacy =~ P1 + P2 + P3 + P4
  Technostress =~ T1 + T2 + T3 + T4
'
fit4 <- lavaan::cfa(model4, data = dat, std.lv = TRUE)
```

The omitted loading arrows are fixed zero, the four latent variances set the factor metrics, and the six pairwise factor covariances are free. With 16 continuous indicators, the covariance ledger is therefore

$$u = \frac{16(17)}{2} = 136, \qquad t = 16 \text{ loadings} + 16 \text{ residual variances} + \frac{4(3)}{2} \text{ factor correlations} = 38, \qquad df = 98.$$

This transfer matters because visual density can conceal a counting error. Four double-headed arrows would omit two factor correlations; counting 16 drawn loading slots as 16 *free* loadings would be wrong under marker scaling but correct under the fixed-factor convention used here. Always state the metric before reading freedom from a diagram.

3 Free, fixed, and constrained parameters

3.1 Different restrictions do different jobs

A free parameter is estimated from the data under the other restrictions. A fixed parameter is assigned a known constant. An equality constraint requires two or more parameter entries to share a value; it does not necessarily assign that value in advance. A bound restricts the allowable range. A defined parameter is a computed function of fitted parameters and does not add a new free parameter.

Entry	Status	Statistical role	Reason to state
$\lambda_{11} = 1$	Fixed	Factor metric	Marker choice
λ_{21}	Free	Target loading	Indicator assignment
$\lambda_{22} = 0$	Fixed	Excluded secondary loading	Content/design restriction
$\lambda_{21} = \lambda_{31}$	Equality	Shared loading value	Equality on a stated scale
$\theta_{23} = 0$	Fixed	Residual relation excluded	Local measurement assumption
ϕ_{12}	Free	Factor association	Correlation permitted
$\theta_{jj} > 0$	Admissibility condition	Positive residual variance	Statistical parameter space
$\lambda_{21} - \lambda_{31}$	Defined	Loading difference	Derived estimand, not extra freedom

Fixing one loading to one selects a unit. Fixing a secondary loading to zero excludes a relation. They are both constants in software, but only the latter ordinarily adds a substantive covariance restriction when the former replaces another valid scale normalization.

"Unconstrained" is also relative. A free loading is estimated within a model whose other loadings, residual structure, and factor variances may be strongly restricted. Its estimate is not model-free evidence about that indicator.

3.2 Change one loading restriction

Allow Y_3 to load on factor 2 as well:

$$Y_{i3} = \nu_3 + \lambda_{31}\eta_{i1} + \lambda_{32}\eta_{i2} + \epsilon_{i3}.$$

One formerly zero matrix entry becomes free. The new coefficient affects the variance of Y_3 and its covariances with all indicators connected to either factor. It does not merely repair the covariance of one chosen pair. The coefficient's interpretation is conditional on factor 1, just as an oblique EFA pattern coefficient was conditional on the other factors.

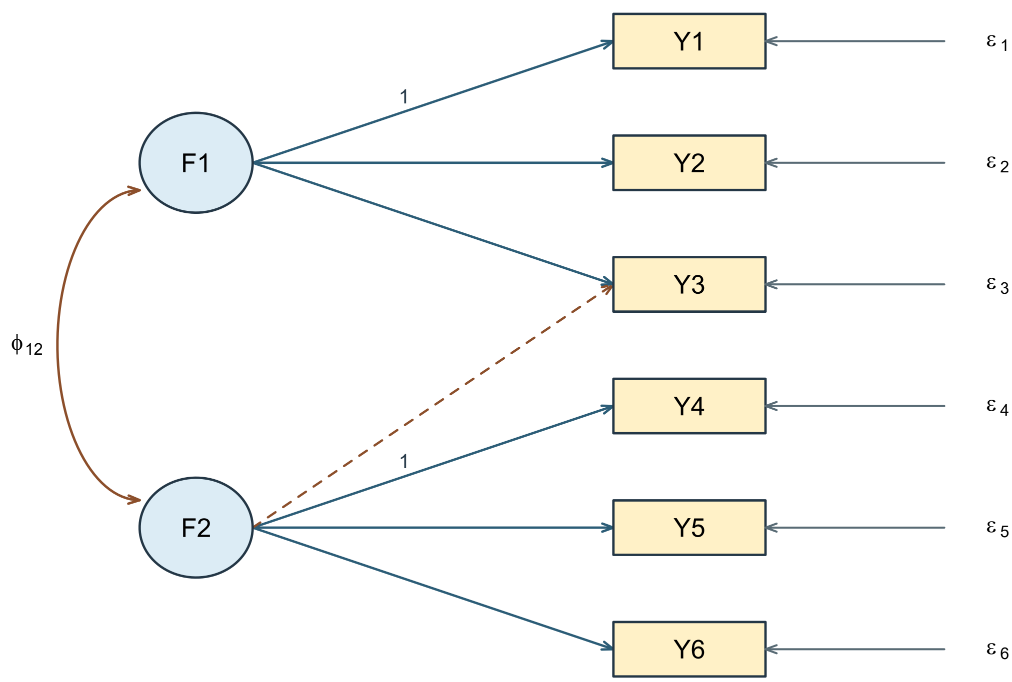


Figure 4: A single additional loading changes several implied moments. The dashed path is lambda32.

3.3 Change one residual relation instead

Alternatively, retain the original loading matrix but free $\theta_{23} = \mathrm{Cov}(\epsilon_2, \epsilon_3)$. Then only the $(2, 3)$ and $(3, 2)$ entries of the residual matrix change. This can represent item-specific shared wording, a common task, or another source of association outside the two modeled factors. Its meaning must be justified; "the software suggested it" is not a measurement explanation.

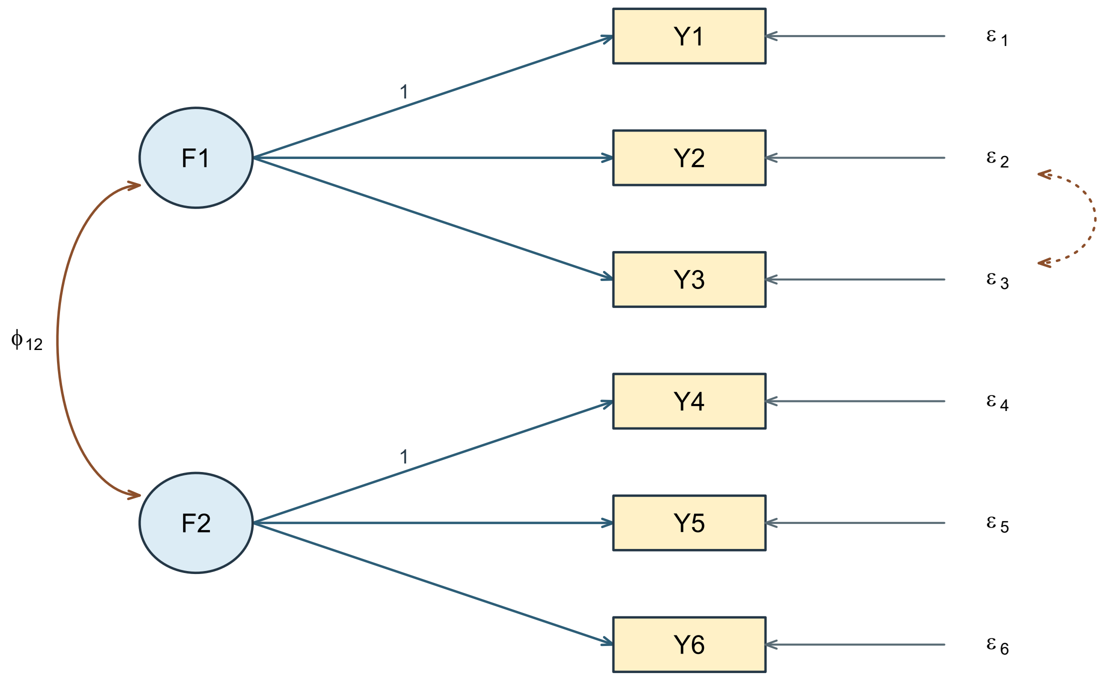


Figure 5: A residual covariance is a different model change from a cross-loading. The dotted relation connects e2 and e3.

Both changes add one parameter, but they generally produce different sets of implied covariance matrices. Equal df therefore does not imply equivalent models. When two modifications improve the same observed pair, their consequences for the other pairs help distinguish them. Local diagnosis depends on that difference.

3.4 Equality requires a common metric

The restriction $\lambda_{21} = \lambda_{31}$ states equal raw changes in Y_2 and Y_3 for a one-unit change in their common factor. If Y_3 is rescaled from points to tens of points, the equivalent restriction must be transformed. Repeating the old equality on rescaled data tests a different hypothesis.

Equality of loadings is not equality of residual variances, intercepts, or item distributions. Equal loadings with unequal residual variances allow different precision. Stronger parallel-measurement conditions require additional restrictions. A short-form scale cannot be justified simply by retaining the largest estimated loadings; content coverage and intended score use also change when items are removed.

4 Give latent variables a metric

4.1 Marker scaling and fixed-factor scaling

Latent variables have no observed unit until we assign one. Under marker scaling, $\lambda_{11} = 1$ means that a one-unit change in η_1 contributes one unit to Y_1 through its loading relation. It does not set the factor variance to one or make the marker error-free.

Under fixed-factor scaling, set $\phi_{11} = \phi_{22} = 1$ and estimate all six target loadings. A factor unit is then one model-standard-deviation unit in this single-group CFA. The indicators retain their raw units. In `lavaan`, `std.lv=TRUE` requests this scale choice. It is not a command to standardize every observed column.

Quantity	Marker scaling	Fixed-factor scaling
Marker loadings	Fixed at one	Free
Factor variances	Free	Fixed at one
Free target loadings	4	6
Free factor variance/covariance entries	3	1
Free residual variances	6	6
Total free covariance axes	13	13

One must also choose factor signs. With unit factor variances, changing the sign of a factor and every loading on it leaves the covariance unchanged. A declared positive orientation for central indicators makes reported signs interpretable; a negative factor does not represent a different fit merely because its column points the other way.

4.2 Derive the reparameterization

For an invertible diagonal $m \times m$ matrix $\mathbf{D}$, define

$$\eta_i^* = \mathbf{D}\eta_i, \qquad \Lambda^* = \Lambda\mathbf{D}^{-1}, \qquad \Phi^* = \mathbf{D}\Phi\mathbf{D}^{\mathsf{T}}.$$

Then

$$\Lambda^*\Phi^*(\Lambda^*)^{\mathsf{T}} = \Lambda\Phi\Lambda^{\mathsf{T}}.$$

For positive factor variances, choosing $\mathbf{D} = \operatorname{diag}(\phi_{11}^{-1/2}, \phi_{22}^{-1/2})$ makes the transformed factor variances one. The residual matrix is unchanged. If latent means are modeled, transform $\alpha^* = \mathbf{D}\alpha$ as well.

Use the following controlled values for hand calculation:

$$\Lambda = \begin{bmatrix} 1 & 0 \\ .8 & 0 \\ 1.2 & 0 \\ 0 & 1 \\ 0 & 1.1 \\ 0 & .9 \end{bmatrix}, \quad \Phi = \begin{bmatrix} 4 & 1.2 \\ 1.2 & 9 \end{bmatrix}, \quad \Theta = \mathrm{diag}(2, 3, 4, 3, 4, 5).$$

The transformation divides factor 1 by 2 and factor 2 by 3. Accordingly, the transformed loading columns multiply by 2 and 3. The covariance 1.2 becomes a correlation $1.2/(2 \times 3) = .20$. The transformed primary loading of Y_2 is 1.6, but its implied variance has not changed.

Table 1
Controlled values under two metrics; these are stipulated parameters, not sample estimates.

Item	Marker loading	Fixed factor loading	Residual variance	Implied variance
Y1	1.000	2.000	2.000	6.000
Y2	0.800	1.600	3.000	5.560
Y3	1.200	2.400	4.000	9.760
Y4	1.000	3.000	3.000	12.000
Y5	1.100	3.300	4.000	14.890
Y6	0.900	2.700	5.000	12.290

This is a reparameterization of the same covariance model, not evidence that marker and fixed-factor methods estimate different constructs. Equivalence can fail if a constraint is not transformed with the metric, if a required marker loading is zero, or if a factor variance is on an inadmissible boundary. Multigroup means and nonlinear models need additional care. A convenient single-group result should not be generalized without checking those conditions.

4.3 Changing observed units is a different operation

Latent rescaling leaves the observed variables untouched. Changing an indicator from raw points to tens of points changes the numerical record itself. Let

$$\mathbf{y}_i^* = \mathbf{b} + \Delta \mathbf{y}_i,$$

where Δ is a nonsingular diagonal matrix of observed-unit conversion factors. The matching model is

$$\begin{aligned}\nu^* &= \mathbf{b} + \Delta\nu, \quad \Lambda^* = \Delta\Lambda, \\ \Theta^* &= \Delta\Theta\Delta^\top, \quad \Sigma^* = \Delta\Sigma\Delta^\top.\end{aligned} \tag{1}$$

Thus multiplying one indicator by 10 multiplies its intercept and loading by 10, its variance and residual variance by 100, and its covariance with an unchanged indicator by 10. A fixed shift changes its intercept and mean but not its covariance. Nothing is unidentified in this operation: the same observed information is expressed in different units, and every affected model quantity must change coherently.

Let $\mathbf{D}_y = \operatorname{diag}(\Sigma)$ contain the positive observed variances. The corresponding correlation model is

$$\begin{aligned}\mathbf{R} &= \mathbf{D}_y^{-1/2}\Sigma\mathbf{D}_y^{-1/2} \\ &= \widetilde{\Lambda}\Phi\widetilde{\Lambda}^\top + \widetilde{\Theta},\end{aligned} \tag{2}$$

where $\widetilde{\Lambda} = \mathbf{D}_y^{-1/2}\Lambda$ and $\widetilde{\Theta} = \mathbf{D}_y^{-1/2}\Theta\mathbf{D}_y^{-1/2}$. This standardizes the indicators, not the factors. With continuous ML and a consistently transformed model, observed-unit rescaling does not change fit. Repeating an untransformed numerical equality constraint after changing units, however, tests a different model.

4.4 Three kinds of coefficient output

For a loading λ_{ja}, latent-standardized and fully standardized versions are, respectively,

$$\lambda_{ja}^{(\mathrm{lv})} = \lambda_{ja}\sqrt{\phi_{aa}}, \qquad \lambda_{ja}^{(\mathrm{all})} = \lambda_{ja}\frac{\sqrt{\phi_{aa}}}{\sqrt{\sigma_{jj}}}.$$

The first expresses a factor change in standard-deviation units while retaining the indicator's raw unit. The second standardizes both. In our fixed-factor CFA, raw and latent-standardized loadings coincide, but fully standardized loadings generally differ.

Standardizing output after fitting does not impose equal indicator variances during estimation. Moreover, the uncertainty of a standardized coefficient depends on the estimated quantities used in the transformation. Dividing a raw standard error by a displayed rounded standard deviation is not generally the correct standard error for the fully standardized coefficient.

A loading greater than one in raw units is unremarkable. Even a standardized pattern coefficient needs to be interpreted in its multivariate context; admissibility concerns the full covariance decomposition. For a single-loading item with uncorrelated residuals, the squared fully standardized loading is its communality. For a cross-loading item, use the full quadratic form, not a single square.

5 Trace restrictions into implied moments

5.1 Within-factor and between-factor covariance

From the controlled marker-scaled model,

$$\begin{aligned}\mathrm{Var}(Y_2) &= \lambda_{21}^2\phi_{11} + \theta_{22},\\ \mathrm{Cov}(Y_2, Y_3) &= \lambda_{21}\lambda_{31}\phi_{11},\\ \mathrm{Cov}(Y_2, Y_5) &= \lambda_{21}\lambda_{52}\phi_{12},\\ \mathrm{Cov}(Y_5, Y_6) &= \lambda_{52}\lambda_{62}\phi_{22}.\end{aligned}$$

The numerical values are 5.56, 3.84, 1.056, and 8.91. Each expression has a reason. Y_2's variance contains common and residual variance. Y_2 and Y_3 share factor 1. Y_2 and Y_5 are connected through the factor covariance. There is no residual cross-term because the base residual matrix is diagonal.

The diagonal and off-diagonal equations do different work during estimation. The covariances inform relations among loadings and factor covariances; the variances also constrain residual variances. An implausibly large common part may force a residual variance toward zero or below it. A Heywood case is therefore a symptom of the whole decomposition, not a harmless cosmetic warning.

Table 2

Full implied covariance matrix from the controlled six-indicator model.

Indicator	Y1	Y2	Y3	Y4	Y5	Y6
Y1	6.000	3.200	4.800	1.200	1.320	1.080
Y2	3.200	5.560	3.840	0.960	1.056	0.864
Y3	4.800	3.840	9.760	1.440	1.584	1.296
Y4	1.200	0.960	1.440	12.000	9.900	8.100
Y5	1.320	1.056	1.584	9.900	14.890	8.910
Y6	1.080	0.864	1.296	8.100	8.910	12.290

```
Sigma <- Lambda %*% Phi %*% t(Lambda) + Theta
Sigma[2, 3]
Sigma[2, 5]
cov2cor(Sigma)
```

The final line changes covariance units to correlations; it does not re-estimate the model. In a fitted model, compare the implied covariance to the sample covariance using matching divisors and variable order.

5.2 What the cross-loading adds

Now add λ_{32} to the third indicator's row. Its variance is

$$\sigma_{33} = \lambda_{31}^2\phi_{11} + \lambda_{32}^2\phi_{22} + 2\lambda_{31}\lambda_{32}\phi_{12} + \theta_{33}.$$

The covariances with Y_2 and Y_5 become

$$\begin{aligned}\sigma_{23} &= \lambda_{21}(\lambda_{31}\phi_{11} + \lambda_{32}\phi_{12}),\\ \sigma_{35} &= \lambda_{52}(\lambda_{31}\phi_{12} + \lambda_{32}\phi_{22}).\end{aligned}$$

If $\lambda_{32} = .25$ while the other controlled parameters stay fixed, σ_{23} increases by $.8 \times .25 \times 1.2 = .24$, whereas σ_{35} increases by $1.1 \times .25 \times 9 = 2.475$. The unequal changes follow from the pathways and factor variances, not a software accident.

The six-indicator example used for estimation below is simulated from exactly this population, with the additional .25 loading included and $n = 500$. Its values are hypothetical: they are generated from the parameters stated above rather than collected from anyone, which is what allows the fitted results to be checked against known truth. It is a separate teaching example and has nothing to do with the twelve-indicator samples. The base six-indicator fit illustrates parameterization; whether that model is adequate is a separate question.

5.3 What a residual covariance adds

Instead, in the base model, adding $\theta_{23} = .50$ changes σ_{23} from 3.84 to 4.34 and leaves all other unique covariance entries unchanged when the remaining parameters are held fixed. This contrasts sharply with the cross-loading's multi-pair changes.

After refitting, other estimates will generally change too. The controlled comparison holds them fixed only to expose the direct algebraic consequences of the specification. Do not confuse that hand calculation with the total change after an optimizer re-estimates every free parameter.

5.4 Means are a separate set of moments

In covariance-only analysis there are $p(p+1)/2$ observed moments. Adding a mean structure adds p more:

$$u_{\text{cov}} = \frac{p(p+1)}{2}, \qquad u_{\text{mean+cov}} = \frac{p(p+1)}{2} + p.$$

If all indicator intercepts are free and latent intercepts are fixed to zero, the extra p mean moments are matched by p extra parameters. The covariance restrictions and df are unchanged.

This is different from testing structured mean differences or constraining intercepts across groups, topics that will become important later.

Freeing every indicator intercept and every latent intercept at once does not identify a latent origin. For any vector $\mathbf{c}$,

$$\alpha^* = \alpha + \mathbf{c}, \qquad \nu^* = \nu - \Lambda\mathbf{c}$$

produce the same observed means. Location restrictions are needed in addition to scale restrictions. Fixing factor variances to one does not remove this location ambiguity.

6 Can the parameters be recovered?

6.1 Identification is about the parameter-to-moment mapping

Identification is a property of the mapping from parameters to moments, and it has to be established for each specification rather than assumed; Bollen (1989) develops the general treatment and Jöreskog (1969) the confirmatory factor case. After accounting for scale and sign conventions, global identification requires that

$$\Sigma(\vartheta_1) = \Sigma(\vartheta_2) \quad \Longrightarrow \quad \vartheta_1 = \vartheta_2$$

for admissible parameter vectors in the declared model. Local identification requires uniqueness in a neighborhood. Generic identification permits exceptional parameter sets, such as special zero loadings, where recoverability fails. For models with means, replace the covariance mapping by the joint mean-and-covariance mapping.

Identification is not the existence of an exact solution to every sample covariance matrix. An identified overrestricted model usually cannot reproduce arbitrary sample moments exactly; estimation finds the closest admissible representation under a stated discrepancy. Conversely, multiple parameter vectors can reproduce the same moments even if an optimizer prints only one of them.

6.2 Count moments and free parameters

Counting begins by choosing one coherent moment representation. For p continuous indicators, a covariance matrix supplies $p(p+1)/2$ distinct variances and covariances. A correlation matrix supplies only $p(p-1)/2$ freely varying off-diagonal entries because its p diagonal entries equal one. Adding an unrestricted observed mean vector contributes another p moments.

For six indicators this gives

Moment representation	Distinct information
Covariances, including variances	$6(7)/2 = 21$
Correlations, with unit diagonals fixed	$6(5)/2 = 15$
Covariances plus observed means	$21 + 6 = 27$

In covariance form, the fixed-factor two-factor simple-structure model has six loadings, one factor correlation, and six residual variances: $t = 13$ and $df = 21 - 13 = 8$. In correlation form, the six unit-diagonal equations determine the six residual variances, leaving seven free parameters and the same $df = 15 - 7 = 8$. Subtracting all 13 covariance-model parameters from 15 correlations would count standardization twice.

For the marker-scaled six-indicator base model,

$$u = \frac{6(7)}{2} = 21, \qquad t = 4 + 3 + 6 = 13, \qquad df = 21 - 13 = 8.$$

Adding one cross-loading yields $t = 14$, $df = 7$. Adding one residual covariance instead gives the same count but a different model. Adding both gives $t = 15$, $df = 6$. A single independent equality constraint on two otherwise free loadings reduces the free-axis count by one. Merely giving a free parameter a label does not change the count.

Six-indicator specification	Loading axes	Factor covariance axes	Residual axes	Candidate df
Marker-scaled base	4	3	6	8
Fixed-factor base	6	1	6	8
Base plus one cross-loading	5	3	6	7
Base plus one residual covariance	4	3	7	7
Base with one loading equality	3	3	6	9

The count $df = u - t$ is necessary bookkeeping, not sufficient proof. The term “observed moments” is preferable to “number of observations” here: 21 unique covariance entries are not 21 people. Increasing the sample size improves precision under suitable conditions but does not create new types of covariance moments or repair structural nonidentification.

The same bookkeeping clarifies why a displayed unrestricted EFA loading matrix contains more entries than the model has distinguishable parameters. With six indicators and two unit-variance orthogonal factors, there are 12 raw loading entries and six residual variances. A two-factor orthogonal rotation has $2(2-1)/2 = 1$ continuous degree of freedom and changes the loading display without changing its reproduced covariance. The generic dimension is

$$t_{\mathrm{EFA}} = 12 + 6 - \frac{2(2-1)}{2} = 17, \qquad df = 21 - 17 = 4. \tag{3}$$

This subtraction counts distinct covariance-model axes at regular, full-rank parameter points. It does not declare any substantively preferred loading to be known, and it does not replace an identification analysis.

6.3 A recoverable block and a counterexample

For a single marker-scaled factor with three indicators and diagonal residuals,

$$\sigma_{12} = \lambda_2\phi, \quad \sigma_{13} = \lambda_3\phi, \quad \sigma_{23} = \lambda_2\lambda_3\phi.$$

When the relevant covariances are nonzero, these imply

$$\phi = \frac{\sigma_{12}\sigma_{13}}{\sigma_{23}}, \quad \lambda_2 = \frac{\sigma_{23}}{\sigma_{13}}, \quad \lambda_3 = \frac{\sigma_{23}}{\sigma_{12}}.$$

The diagonal equations then determine residual variances. These expressions explain generic recoverability for this simple block, subject to admissibility. If a denominator vanishes, the reasoning must be revisited; a generic result does not cover every exceptional configuration.

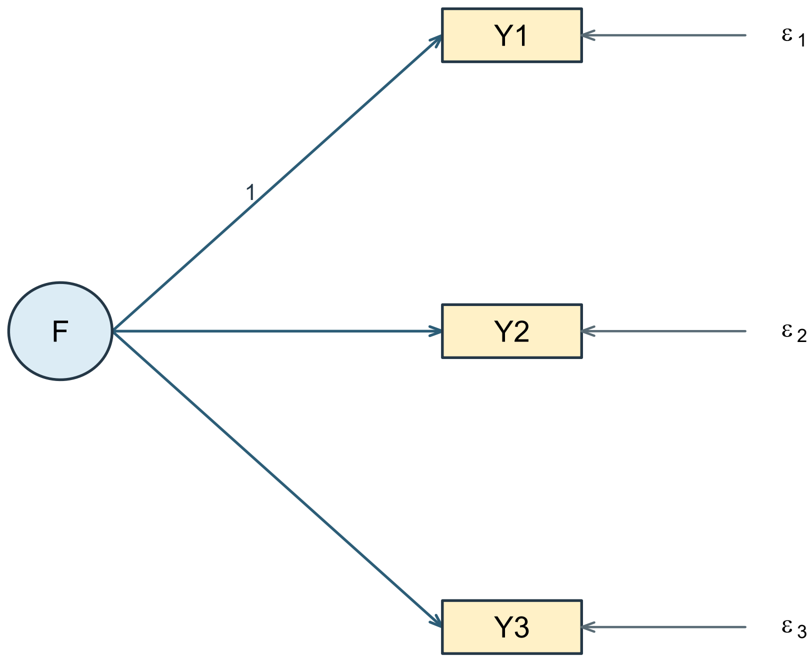


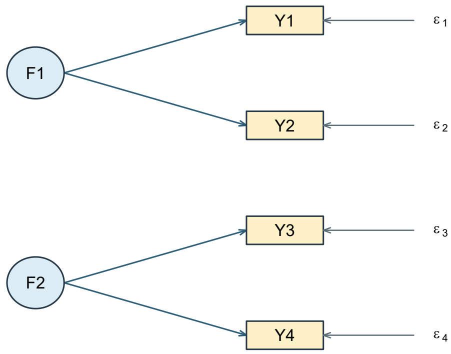
Figure 6: A marker-scaled three-indicator block. The three off-diagonal covariances can recover the factor variance and two non-marker loadings at regular, nonzero parameter points.

Now consider two *orthogonal*, isolated two-indicator factors, with unit factor variances. Each block has two free loadings and two residual variances but only three covariance moments. Across both blocks there are ten available covariance entries and eight free parameters, giving a superficially favorable $df = 2$. Yet the cross-block covariances are all fixed to zero and provide no information to separate the two loadings within either block. For one block,

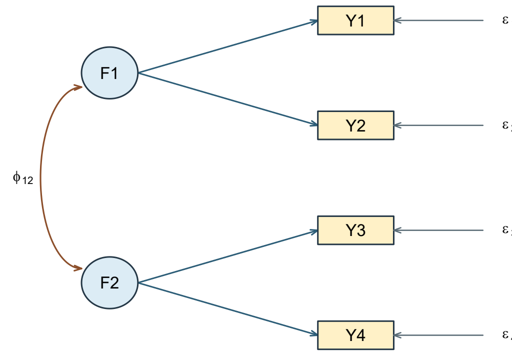
$$\sigma_{12} = \lambda_1 \lambda_2, \quad \theta_{11} = \sigma_{11} - \lambda_1^2, \quad \theta_{22} = \sigma_{22} - \lambda_2^2.$$

A range of choices for λ_1 can yield the same moments by adjusting λ_2 and both residual variances, while retaining positive variances. The whole-model count did not reveal the local deficiency. Additional indicators or justified restrictions can help; adding an arbitrary constraint solely to make software run changes the measurement model and needs a rationale.

The diagram pair makes the missing information visible. In the isolated model, all four cross-block covariances are fixed to zero, so the positive whole-model df do not help identify either two-indicator block. Allowing a nonzero factor covariance connects the blocks: cross-block moments then contain products of loadings from both factors. The connected model is generically identified under regular nonzero loadings and $\phi_{12} \neq 0$; the argument degenerates when that connection is zero.



(a) Isolated orthogonal blocks



(b) Connected correlated blocks

Figure 7: Two-indicator blocks can have the same whole-model count but different recoverability because the covariance edge changes which moments carry loading information.

6.4 Restriction is what replaces rotation

It is worth being explicit about why identification becomes a live question here rather than earlier. An exploratory model is unidentified by construction: for any nonsingular $\mathbf{T}$, the pair $\Lambda\mathbf{T}^{-1}$ and $\mathbf{T}\Phi\mathbf{T}^{\top}$ reproduces the same covariance, so a whole continuous family of loading matrices is observationally equivalent, and rotation selects one member of it by a stated criterion.

A confirmatory model does not select from that family; it removes it. Once enough loadings are fixed at zero, no nontrivial $\mathbf{T}$ maps the pattern onto itself - a rotation that preserved the zeros would have to be diagonal, and the scaling convention then fixes what remains up to sign. That is the trade. Exploratory analysis buys freedom from prior commitment at the cost of never identifying its axes; confirmatory analysis buys identified parameters at the cost of having to defend every zero.

Because the zeros do the work, identification now depends on *where* they are, not only on how many there are. Two sufficient conditions cover most applied specifications (Bollen, 1989; Brown, 2006), and both assume a diagonal residual covariance matrix and a scale fixed for each factor:

- **Three indicators per factor.** A factor with at least three indicators that load on it alone is identified, whether or not it correlates with anything else. Section 6.3 did this case explicitly.
- **Two indicators per factor.** A factor with only two exclusive indicators is identified when it is *connected* to the rest of an identified measurement model: it must have a nonzero covariance with at least one other identified factor, and the parameter values must be regular, meaning that the cross-block covariances carrying the information are

not themselves zero. The orthogonal counterexample above fails exactly this condition - the connecting covariances were fixed at zero, so no information reached the block.

Stated with those conditions attached, both are sufficient for generic identification; they are not necessary, and models that violate them may still be identified. The second rule is the more fragile of the two, because it depends on a quantity that is estimated rather than assumed. If the connecting correlation is genuinely zero in the population, the rule simply does not apply. If it is nonzero in the population but estimated near zero in a particular sample, the model remains identified in principle while behaving in estimation as though it were not. That is the situation the next section calls weak identification, and its symptoms are unstable estimates and enormous standard errors rather than an error message.

6.5 Structural, weak, and numerical problems

Structural nonidentification is built into the parameter mapping. Weak identification describes a formally recoverable model whose parameters are poorly distinguished in a particular configuration. Small loadings, nearly proportional loading columns, or factor correlations close to one can make very different parameter vectors produce nearly indistinguishable moments. Numerical difficulty is another issue: unsuitable starts or badly scaled variables can impede an optimizer even when a well-behaved optimum exists.

Check	Warning	What it warrants
Moment count	More free parameters than moments	Reconsider specification
Metric and sign	No normalization or inconsistent orientation	Specify units and signs
Local connections	Isolated insufficient indicator block	Examine block equations
Loading rank	Redundant factor patterns	Examine distinct common directions
Factor covariance	Near singularity	Investigate weak separation
Residual variances	Zero or negative estimates	Inspect admissibility and misspecification
Information/SEs	Singular matrix or unstable large SEs	Check recoverability and numerical behavior
Starts and gradients	Different optima or nonzero gradient	Resolve optimization before interpretation

A numerical derivative matrix provides a local diagnostic. If $\mathbf{s}(\vartheta)$ stacks the distinct implied moments, its Jacobian is $\Delta = \partial\mathbf{s}/\partial\vartheta^{\mathsf{T}}$. Full column rank at a regular point supports local identification; it does not prove global uniqueness or uniform stability throughout the parameter space. Students need not derive this matrix to understand what the diagnostic adds beyond counting.

The SLE primary fit has numerical Jacobian rank 29 for 29 free parameters. That is a useful local check of the actual fitted candidate. The three blocks also have several pure primary indicators, so the two retained cross-loadings do not leave every factor's orientation open as in unrestricted EFA.

6.6 Nested, nonnested, and equivalent models

A restricted model is nested in another if every covariance matrix it can generate belongs to the larger model's covariance set. Show the restrictions that produce that inclusion; a parameter-count comparison alone is not enough. The base six-indicator model is nested in its cross-loading extension by setting $\lambda_{32} = 0$.

Two models can have the same df and different covariance sets, as with an alternative indicator assignment. They can also have different-looking parameters but identical covariance sets, as with the regular marker and fixed-factor reparameterizations. Observational equivalence limits what covariance data can distinguish. It does not establish that higher-order, bifactor, and correlated-residual models are generally equivalent; each such claim requires an exact mapping and conditions.

One deliberately constrained example shows what an exact mapping looks like. For two positively correlated first-order factors, fix both second-order loadings to one, set $\mathrm{Var}(G) = \phi_{12}$, and set the two first-order disturbance variances to $\phi_{11} - \phi_{12}$ and $\phi_{22} - \phi_{12}$. When these variances are nonnegative, the second-order parameterization reproduces exactly the same 2×2 factor covariance matrix. Without the two extra scale/equality restrictions, two first-order factors do not by themselves identify a second-order factor. The substantively more useful cases have three or more first-order factors.

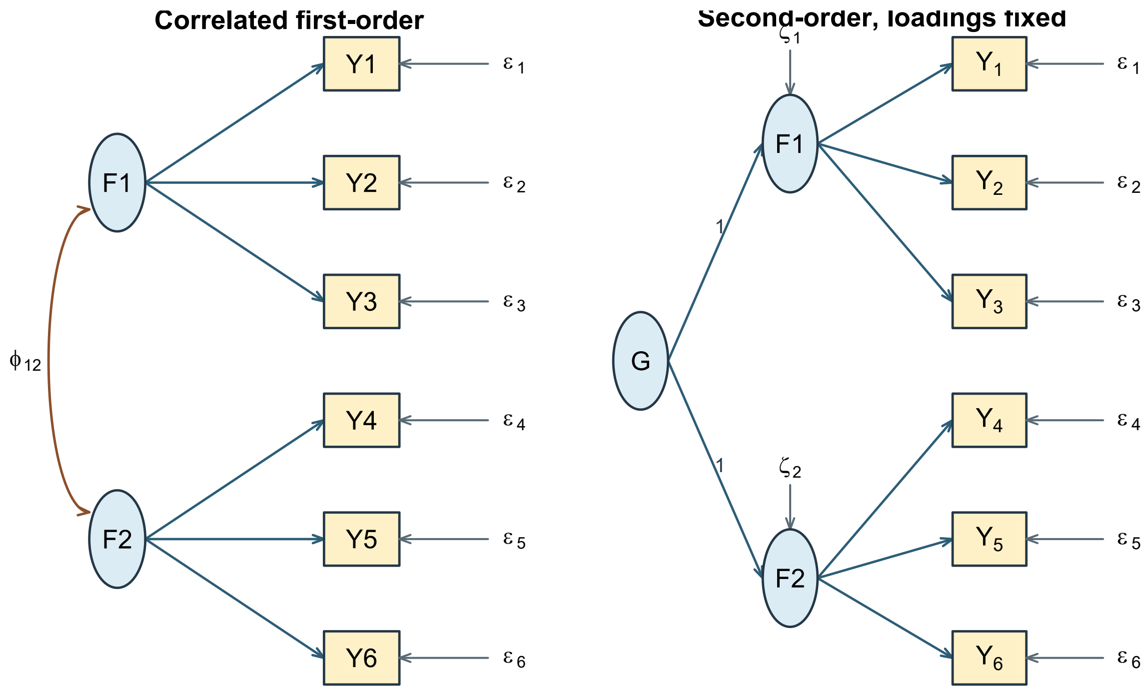


Figure 8: A correlated two-factor model and one constrained second-order reparameterization. The equivalence requires the stated fixed loadings and admissible disturbance variances.

These distinctions prepare comparison. They do not yet say which candidate is adequate or which restriction the data oppose.

7 What estimation does

7.1 Minimize a stated discrepancy

Let $\mathcal{P}$ be the admissible parameter space and $\mathbf{S}$ the sample covariance under a declared divisor. Covariance estimation can be written

$$\hat{\vartheta} = \arg\min_{\vartheta \in \mathcal{P}} F\{\mathbf{S}, \Sigma(\vartheta)\}.$$

For normal-theory ML, a conventional covariance discrepancy is

$$F_{\mathrm{ML}} = \log|\Sigma(\vartheta)| + \mathrm{tr}\{\mathbf{S}\Sigma(\vartheta)^{-1}\} - \log|\mathbf{S}| - p.$$

The determinant summarizes multivariate dispersion; the trace term compares the sample covariance to the model's inverse-covariance weighting. The last two terms depend only on the input matrix and dimension. They normalize the discrepancy to zero when $\Sigma = \mathbf{S}$, because $\mathrm{tr}(\mathbf{S}\mathbf{S}^{-1}) = p$. The dimension term is the number of observed variables, not the number of indicators plus factors.

At a proper optimum, numerical changes in the free parameters cannot materially lower the objective under the specified constraints. That is an optimization statement. It does not imply small absolute discrepancy, plausible parameter meaning, or a unique substantive model.

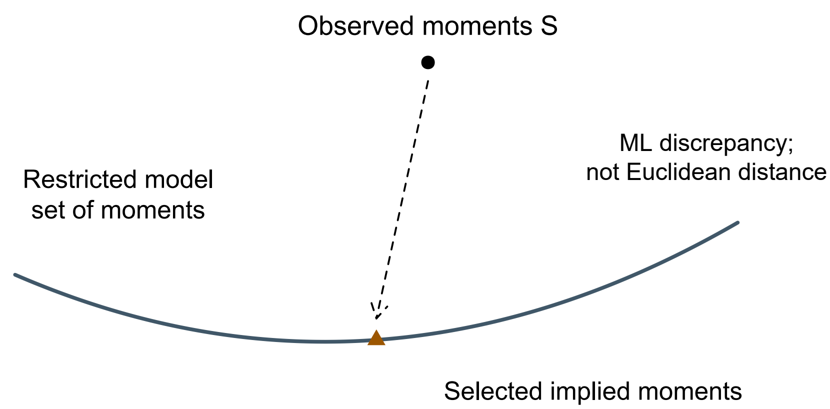


Figure 9: Conceptual estimation geometry. The curve represents a restricted set of moments, not the literal dimension or shape of this CFA.

7.2 Normality belongs to the likelihood contract

The factor covariance equation follows from linearity, finite second moments, and the stated zero covariances. It does not require normality. Conventional normal-theory ML standard errors and tests additionally use sampling and distributional regularity. Where the departure is nonnormality of the observed variables under independent sampling, scaled corrections to the test statistic and standard errors are the standard remedy (Satorra & Bentler, 1994). They do not address dependence between cases, clustering, or selection, each of which needs its own treatment. Independently distributed normal factors and normal residuals give a jointly normal observed vector in this linear model. Marginally normal histograms alone do not establish multivariate normality or independent sampling.

Sample B consists of 360 complete, independently simulated continuous records. There is no silent missing-case removal, imputation, weighting, clustering adjustment, or ordinal recoding. The raw indicators retain the fixed process-index units. The fitted settings are explicit:

```
sle <- read.csv("sle_continuous.csv")

items <- c(paste0("P", 1:4),
           paste0("S", 1:4),
           paste0("B", 1:4))

dat <- sle[sle$sample == "B", items]

model <- '
  Planning =~ p1*P1 + p2*P2 + P3 + P4
  Strategy =~ S1 + S2 + S3 + S4 + P4
  Belonging =~ B1 + B2 + B3 + B4 + S4
'
fit <- lavaan::cfa(model, data = dat[items],
                   estimator = "ML", likelihood = "normal",
                   std.lv = TRUE, meanstructure = FALSE,
                   missing = "listwise", se = "standard",
                   test = "standard", information = "expected")
```

Sample B is the confirmation record in the released `sle_continuous.csv`; the exploratory chapter's data appendix gives the codebook and sample manifest. The syntax above matches the fixed candidate exactly, so the block runs as printed against the distributed file. Before fitting, the chapter source checks the released file against the declared population: the column names and order, the row count for each sample, the case identifiers, and every indicator value. Numerically coded columns alone are not sufficient evidence of a continuous measurement scale.

Categorical indicators are covered elsewhere. Categorical group labels and dummy predictors remain available for later multigroup and MIMIC models. Later SEM chapters address robust ML inference for nonnormality and FIML for missing continuous data under suitable assumptions; neither is a categorical-measurement switch.

7.3 Record the covariance divisor

The normal-likelihood convention uses

$$\mathbf{S}_N = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})^\mathsf{T} = \frac{N-1}{N}\mathbf{S}_{N-1}.$$

R's `cov()` uses $N-1$; `lavaan` performs the normal-likelihood rescaling internally. Do not pre-rescale raw observations to compensate. When working from a covariance matrix rather than raw data, disclose the supplied divisor and package rescaling options. The estimator documentation distinguishes the normal and Wishart conventions explicitly.

For the normal convention, the model statistic uses N times the covariance discrepancy. The Wishart convention uses an $N-1$ covariance and multiplier. Chapter 2's corrected exploratory statistic is a further distinct finite-sample convention. This chapter records the normal family without interpreting its model test.

7.4 A small mean-structure comparison

The fixed M1-cross has 78 covariance moments, 29 free covariance parameters, and 49 df. Adding twelve free indicator intercepts while keeping latent intercepts zero gives 90 moments, 41 free parameters, and the same 49 df. For the stricter M0-simple model the corresponding counts are 78/27/51 and 90/39/51.

Table 3

Counts verified against actual fitted parameter tables. The mean extension does not constrain the indicator means.

Model	Moments	Free	Df
M1 covariance	78	29	49
M1 free intercepts	90	41	49
M0 free intercepts	90	39	51

The largest difference between the M1 covariance-only and free-intercept implied covariance matrices is 0e+00. The estimated free intercepts reproduce the sample means. This verifies the intended separation of covariance restrictions from an unrestricted mean extension. It does not justify freely estimating latent origins without additional constraints.

8 Read an estimated model before reading its fit

8.1 Convergence and admissibility first

The optimizer reports convergence for the inherited M1-cross model, with no captured fitting warnings. The maximum absolute gradient is 3.5e-07. All residual variances are positive, the factor covariance and implied covariance matrices are positive definite, and standard errors are available. The smallest information-matrix eigenvalue is positive, although its magnitude depends on parameter scaling and is not a universal quality index.

Table 4
Selected estimation gates for the actual Sample B primary fit. Passing these gates does not establish adequate fit.

Check	Passed
Converged	TRUE
No warnings	TRUE
Positive residual variances	TRUE
Positive factor covariance	TRUE
Positive implied covariance	TRUE

If convergence fails, first inspect the stated model, data, scale, and starting values. If convergence succeeds but a variance is negative, the solution is improper for this model. Do not proceed directly to fit-index interpretation or replace the negative value by zero in a report. If an admissible fit has large standard errors or near-perfect factor correlations, examine weak separation and sensitivity. Different warnings imply different investigations.

These checks are reproducible from `lavInspect()` and the parameter table. They are diagnostics at the fitted point; they do not substitute for the earlier structural identification reasoning.

8.2 Read loadings in their stated units

Table 5
Sample B loadings for the frozen M1-cross. Intervals are 95 percent normal-approximation intervals for raw coefficients. Std. all is a point estimate on the fully standardized metric.

Factor	Item	Raw	SE	Lower	Upper	Std.all
Planning	P1	6.304	0.407	5.507	7.102	0.770
Planning	P2	7.810	0.520	6.791	8.830	0.751
Planning	P3	8.407	0.629	7.174	9.640	0.684
Planning	P4	4.734	0.472	3.809	5.658	0.519
Strategy	S1	7.524	0.494	6.556	8.493	0.751
Strategy	S2	6.114	0.422	5.287	6.940	0.724
Strategy	S3	7.860	0.579	6.725	8.996	0.687
Strategy	S4	5.438	0.424	4.607	6.270	0.622
Strategy	P4	3.288	0.460	2.387	4.190	0.361
Belonging	B1	8.619	0.450	7.737	9.502	0.888
Belonging	B2	9.210	0.565	8.102	10.317	0.784
Belonging	B3	4.925	0.407	4.127	5.723	0.616
Belonging	B4	4.695	0.554	3.608	5.781	0.453
Belonging	S4	2.569	0.398	1.790	3.348	0.294

For P1, a one-standard-deviation increase in Planning corresponds to an estimated 6.304 process-index-point change through the specified loading relation. Its standard error is 0.407, and its 95% interval runs from 5.507 to 7.102. The fully standardized loading, 0.770, expresses the same relation relative to P1's model-implied standard deviation. Neither magnitude is a causal effect established by this analysis.

P4 and S4 each have two rows because two loadings were fixed as free before Sample B was inspected. In this restricted CFA, the secondary coefficients are estimated jointly with all other free parameters. They need not equal the rotated EFA coefficients from another sample, another matrix metric, and a less restricted loading model.

The signs and intervals support reading these as positive conditional relations within the specified model. They do not license removing other restrictions without evaluation. A precise coefficient can be estimated inside a model whose broader covariance representation is inadequate.

8.3 Standard errors, intervals, and tests

For a regular interior parameter with null value ϑ_0,

$$z = \frac{\hat{\vartheta} - \vartheta_0}{\mathrm{SE}(\hat{\vartheta})}, \qquad \hat{\vartheta} \;\pm\; 1.96\,\mathrm{SE}(\hat{\vartheta})$$

give a conventional asymptotic test and 95% interval under the stated ML assumptions. The interval concerns repeated-sample coverage of a population parameter under that model. It is not a range containing 95% of individual factor scores or item responses.

Normal approximations deserve particular caution for boundary variances, strongly asymmetric sampling distributions, weak identification, and misspecified distributional assumptions. A fixed parameter has no sampling standard error as a freely estimated quantity: its printed value of one or zero is a specification, not a highly precise discovery. Software may print zero SE and omit its z test for that reason.

Statistical significance also differs from measurement importance. A small loading can be precisely estimated in a large sample; a content-essential loading may be uncertain in a small sample. Deleting items solely because $p > .05$ changes the model and the content represented. That decision needs more than a significance column.

8.4 Factor relations and residual variance

Table 6

Estimated factor correlations under fixed-factor scaling, with raw-metric intervals. Here the raw factor covariance is itself a correlation.

Factor1	Factor2	Correlation	SE	Lower	Upper
Planning	Strategy	0.406	0.058	0.292	0.519
Planning	Belonging	0.289	0.059	0.174	0.405
Strategy	Belonging	0.289	0.060	0.171	0.407

All three correlations are positive and distinct from one in the reported point solution. A correlation is not the percentage of one factor caused by another. It also does not measure the reliability of either factor. Later structural models place restrictions on these associations; measurement estimation alone does not select their direction.

For a single-loading item, the standardized residual variance is θ_{jj}/σ_{jj} and the common proportion is its complement. For a cross-loading item,

$$h_j^2 = \frac{\lambda_j^{\mathsf{T}} \Phi \lambda_j}{\sigma_{jj}}.$$

The numerator includes cross-products involving factor correlations. It is not generally the sum of two squared standardized coefficients. Residual variance can include specific stable influences and random measurement error; the factor model does not automatically separate those components.

8.5 A planned equality test

M3-equal was named in Chapter 2: it constrains the raw Planning loadings of P1 and P2 to equality. Let $\mathbf{R}\vartheta = \mathbf{r}$ represent q independent linear restrictions. The Wald statistic is

$$W = (\mathbf{R}\hat{\vartheta} - \mathbf{r})^{\mathsf{T}} \left[\mathbf{R}\widehat{\mathrm{Var}}(\hat{\vartheta})\mathbf{R}^{\mathsf{T}}\right]^{-1} (\mathbf{R}\hat{\vartheta} - \mathbf{r}).$$

Under suitable regularity, its reference is χ^2_q. For one loading difference, the estimated variance is

$$\mathrm{Var}(\hat{\lambda}_{\mathrm{P1}} - \hat{\lambda}_{\mathrm{P2}}) = \mathrm{Var}(\hat{\lambda}_{\mathrm{P1}}) + \mathrm{Var}(\hat{\lambda}_{\mathrm{P2}}) - 2\,\mathrm{Cov}(\hat{\lambda}_{\mathrm{P1}}, \hat{\lambda}_{\mathrm{P2}}).$$

Ignoring the covariance between estimates can give the wrong test. The syntax labels `p1` and `p2` identify the coefficients:

```
lavaan::lavTestWald(fit, constraints = "p1 == p2")
```

Here $W = 6.911$ on one df, with $p = 0.009$. Under the declared model and normal approximation, the data oppose this particular raw-loading equality. This is not a test that M1 is globally adequate, nor a test of equality of fully standardized loadings. The Wald perspective is one of three views of the same restriction; the restricted-model and likelihood-ratio views belong with model comparison.

9 Build the evaluation transition record

9.1 Carry the exact candidate, not an assumed baseline

The inherited model has twelve indicators, three correlated factors, and two additional loadings. Its fixed-factor covariance ledger is

$$u = 78, \qquad t = 14 \text{ loadings} + 12 \text{ residual variances} + 3 \text{ factor correlations} = 29, \qquad df = 49.$$

The simpler 27-parameter, 51-df model remains M0-simple, a named rival. It is not silently substituted for the inherited model. The all-item one-factor rival, the orthogonal restriction, and the raw-loading equality retain their Chapter 2 definitions. Models keep their M0–M4 identifiers across the three chapters; the separate example models use D1–D4, and a later data-developed revision will use an R identifier.

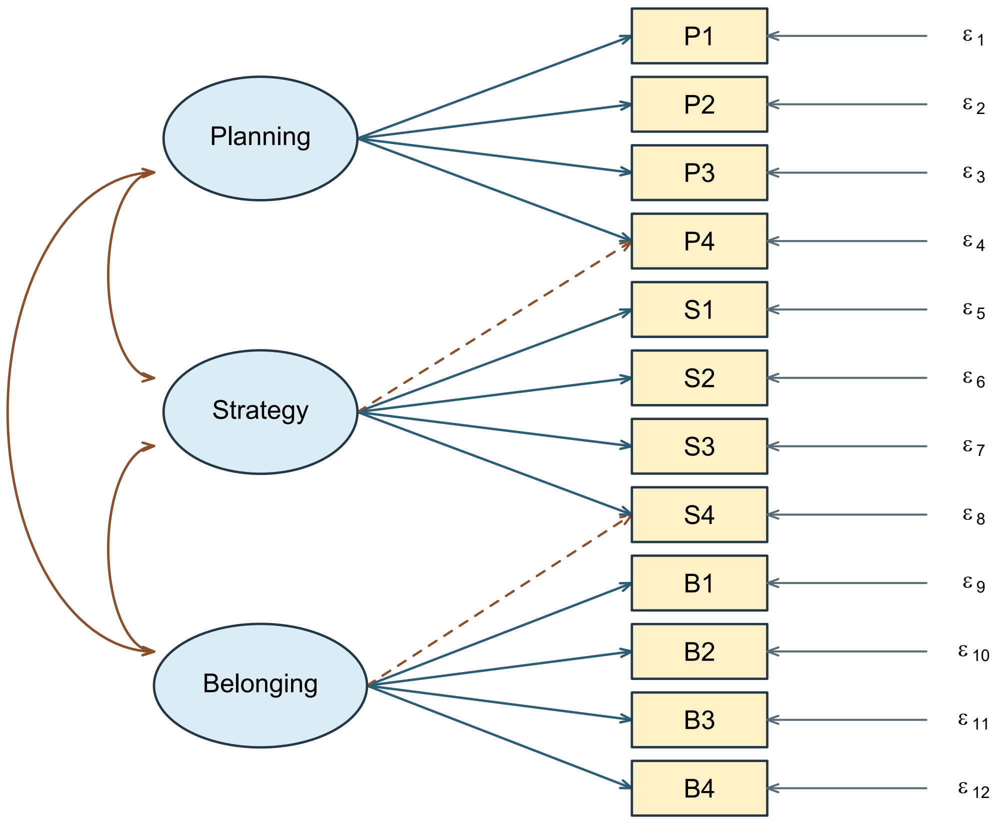


Figure 10: The exact M1-cross structure carried into evaluation, not a redrawn simple-structure substitute.

9.2 Save what another analyst needs

Record component	What the in-memory object carries
Evidence identity	Sample B role, 360 rows, item order and case identifiers
Measurement specification	Fixed syntax, active item list, factors and all rivals
Parameter conventions	Unit factor variances, zero latent location, covariance-only model
Estimation	ML, normal likelihood, expected-information standard errors
Data handling	Complete continuous raw observations; no weights or clustering
Numerical status	Convergence, warnings, gradients, admissibility, local rank
Parameter output	Full free/fixed ledger, estimates, SEs, intervals, standardizations
Fitted moments	Observed $\mathbf{S}_N$ and exact model-implied covariance
Evaluation fields	Conventional test, df, RMSEA/interval, CFI/TLI, SRMR, log likelihood, AIC/BIC
Reproduction	R and package versions, and the model syntax

Nothing here is written to disk. The record is an in-memory object rebuilt from the declarations in this chapter every time it renders, and its baseline is the fitted `lavaan` object itself rather than a reconstruction from rounded coefficients. What the render guarantees is internal consistency - the same declared population, sample role, syntax and settings produce the same fit each time - not agreement with a separately stored file from an earlier session.

The numerical marker/fixed-factor comparison also succeeds for both records. For the SLE fit, the largest absolute difference in implied covariance is 1.4e-04; for the six-indicator example fit it is 4.7e-06. The small numerical tolerances reflect separate optimization, not a theoretical discrepancy between valid scale choices.

Saving evaluation fields is not interpreting them. They are checked for availability and matching conventions here so that evaluation can begin with an intact record. The validation sample has not been analyzed in building it.

10 Summary and reading bridge

Five equations organize the chapter: the scalar measurement equation, the covariance decomposition, the scale transformation, the parameter-to-moment identification mapping, and the discrepancy minimization. Together they distinguish a proposed structure from a recoverable parameterization and a successful numerical estimate.

A disciplined specification check asks whether words, equations, matrices, diagram, and syntax describe the same model. A parameter check then separates free entries, scale normalizations, substantive zeros, equalities, and derived quantities. The count of distinct moments minus free parameters is essential but does not replace local-block or identification reasoning. Mean models need location conventions as well as factor scales.

The Sample B candidate passes the stated estimation and admissibility checks. Its parameters can be read with explicit units and conventional uncertainty. That is the stopping point: **an identified, converged, admissible estimate is necessary for ordinary interpretation, but it is not evidence by itself of adequate model fit.**

The technical route follows the CFA specification tradition developed in Jöreskog (1969), Bollen (1989), and Brown (2006). For software details, the parameter-syntax tutorial and output-extraction guide help connect restrictions and fitted-object components. What follows examines the exact-fit test, fit profile, local residuals, planned comparisons, and the conditions under which a revision would need new evidence.

Exercises

1. **Represent a model.** Write the six-indicator base model in equations, matrices, and syntax. Identify its four free non-marker loadings, two marker constraints, six target zeros, and residual assumptions. Explain why no drawn arrow is causal proof.
2. **Trace a changed relation.** Using the controlled parameter values in Section 4, compute σ_{16} and σ_{36}. Add $\lambda_{32} = .25$ and recompute the affected moments. Contrast adding $\theta_{23} = .50$ instead.
3. **Transform the metric.** For marker-scale factor variances 9 and 4, covariance 1.8, and target loadings $(1, .7, 1.1)$ and $(1, .8, 1.2)$, obtain the fixed-factor loading matrix and factor correlation. Verify a within-factor and a between-factor covariance before and after transformation.
4. **Count carefully.** Evaluate the six-indicator base, its one-cross-loading extension, and its loading-equality restriction. Reconstruct the four-factor, sixteen-indicator count $136 - (16 + 16 + 6) = 98$. Then derive $78/(27 + k)/(51 - k)$ for twelve indicators and k extra free cross-loadings under the stated simple-structure assumptions.
5. **Count is not identification.** Construct two distinct admissible loading/residual-variance parameter vectors for a two-indicator unit-variance factor with $\sigma_{11} = \sigma_{22} = 1$ and $\sigma_{12} = .4$. Explain how an isolated copy of this block can remain unidentified inside a positive-df whole model.
6. **Read uncertainty.** From the Sample B loading table, interpret P3's raw coefficient, SE, interval, and fully standardized coefficient. Explain why its raw coefficient above one is not an improper solution and why the printed raw SE is not the fully standardized SE.

7. **Examine the contract.** An analyst feeds R's `cov()` matrix into another program, leaves its divisor unspecified, fixes marker loadings to one, and compares its test directly with this chapter's raw-data fit. List the settings and invariances that must be checked before interpreting a numerical difference.
8. **Test a restriction.** Explain the P1/P2 raw-loading equality, write its $\mathbf{R}$ row, and identify the covariance term required for the variance of the estimated difference. Why does this test not decide global fit or full parallel measurement?

Appendix: Structured means and parcels

Latent location is not identified by naming a factor

The mean equation $\mu = \nu + \Lambda\alpha$ contains both indicator intercepts and latent intercepts. In a single group, zero latent intercepts with free indicator intercepts are a convenient reference convention. In multigroup work, some intercept restrictions can anchor latent mean differences, but those restrictions carry measurement content. A numerical origin does not arise from labeling one group "control" or one factor "ability." Later invariance chapters develop the required scale and location links.

The covariance-only model is unaffected by translating every observed column by a constant. A structured-mean model need not be. Similarly, rescaling a factor requires transforming its loadings, covariance, and latent intercept; rescaling an observed variable requires transforming its intercept and residual variance as well. An equality imposed after only some of these transformations can silently change the hypothesis.

Parceling changes the observed system

Combining items into parcels before fitting is common and contested; the trade-offs are set out in Little et al. (2002). The algebra below shows why the choice is not cosmetic. Let a vector of parcel scores be $\mathbf{z} = \mathbf{A}\mathbf{y}$, where rows of $\mathbf{A}$ specify item averages or sums. Then

$$E(\mathbf{z}) = \mathbf{A}\mu, \qquad \mathrm{Var}(\mathbf{z}) = \mathbf{A}\Sigma\mathbf{A}^{\mathsf{T}}.$$

If $\mathbf{A}$ reduces dimension, several different item-level covariance patterns can yield the same parcel covariance. Opposing secondary loadings or local dependencies may partly cancel within averages. A well-fitting parcel model therefore does not establish that the item-level restrictions would fit, and its AIC/BIC is not directly comparable to an item-level model with different outcomes.

Parceling can be useful under a justified measurement design, but it is not a generic normality repair or a route around identification and dimensionality questions. Evaluate item structure and preserve the meaning of the resulting score before changing the observed system. No parcels are used in the SLE course record.

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